

Protocol



'Back-to-base' combined hypothermic and normothermic machine perfusion of human donor livers

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Abstract

The shortage of suitable donor organs has resulted in the use of suboptimal, high-risk, extended-criteria donor (ECD) livers, which are at an increased risk of failure after transplantation. Compared with traditional static cold storage, dynamic preservation by ex situ machine perfusion reduces the risks associated with the transplantation of ECD organs. Ex situ machine perfusion strategies differ in timing (that is, speed of procurement and transport), perfusion duration and perfusion temperature. For 'back-to-base' protocols, the donor liver is statically cold stored during transportation to the recipient hospital (the 'base') and then perfused, instead of transporting the liver using a portable perfusion system. While dual hypothermic (8–12 °C) oxygenated machine perfusion (DHOPE) allows safe prolongation of preservation duration and reduces ischemia–reperfusion injury-related complications, including post-transplant cholangiopathy, normothermic machine perfusion (NMP) at 35–37 °C facilitates ex situ viability testing of both liver parenchyma and bile ducts. Here, we describe a clinical protocol for 'back-to-base' combined DHOPE and NMP, linked by a period of controlled oxygenated rewarming (COR), which we call the DHOPE–COR–NMP protocol. This protocol enables restoration of mitochondrial function after static ischemic preservation and minimizes both ischemia–reperfusion and temperature-shift-induced injury during the start of NMP. The NMP phase allows viability assessment before final donor liver acceptance for transplantation. Sequential DHOPE and COR–NMP may reduce the risks associated with transplantation of ECD livers and facilitate enhanced utilization, thereby helping to alleviate the organ shortage.

Key points

- Here, the authors describe a clinical protocol for 'back-to-base' combined dual hypothermic oxygenated machine preservation and normothermic machine perfusion, linked by a period of controlled oxygenated rewarming.
- This protocol minimizes both ischemia–reperfusion and temperature-shift-induced injury during the start of normothermic machine perfusion.

Key references

- van Leeuwen, O. et al. *Ann. Surg.* **270**, 906–914 (2019): <https://doi.org/10.1097/SLA.0000000000003540>
- van Leeuwen, O. et al. *Am. J. Transplant.* **22**, 1658–1670 (2022): <https://doi.org/10.1111/ajt.17022>

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Introduction

Development of the protocol

In most high- and middle-income countries, donation after brain death rates are slowly decreasing, while donation after circulatory death (DCD) rates are increasing¹. Additionally, the average donor age and body mass index continue to rise, as well as the incidence of comorbidities, such as diabetes mellitus, liver steatosis and cardiovascular disease². The increased use of DCD livers for transplantation, together with the reduced quality of these grafts because of severe donor comorbidities, has led to high incidences of severe post-transplant complications, including primary non-function and post-transplant cholangiopathy^{3–7}. Consequently, large numbers of donor livers are declined for transplantation because the risk of primary non-function or post-transplant cholangiopathy is considered too high. It is expected that in the USA, more than 50% of all potentially available donor organs will be declined for liver transplantation by the year 2030².

As a result of the increasing donor liver discard rates and post-transplant complications, the previous gold standard of graft preservation on ice (static cold storage (SCS)) was challenged by a reintroduction of machine perfusion to clinical practice in 2010⁸.

Ex situ machine perfusion strategies differ in timing, duration, perfusion temperature, type of perfusion fluid, single (via the portal vein only) versus dual (via the portal vein and the hepatic artery) perfusion and oxygenation activity. Clinically, the most frequently used strategies are post-SCS, single or dual hypothermic (4–12 °C) oxygenated machine perfusion ((D)HOPE) or normothermic (35–37 °C) machine perfusion (NMP). Machine perfusion protocols may optimize donor liver utilization and reduce post-transplant complications via graft reconditioning, viability assessment, improved preservation and potential therapeutic interventions such as defatting or regenerative therapies.

DHOPE mitigates ischemia–reperfusion injury by restoring mitochondrial function before transplantation^{8–11}. Reintroducing oxygen at low temperatures (8–12 °C) effectively decreases accumulated metabolites in the mitochondria without producing excessive amounts of reactive oxygen species¹². Additionally, ATP production exceeds demand, allowing for replenishment of cellular energy levels¹⁰. The first multicenter international randomized controlled trial comparing end-ischemic DHOPE with SCS demonstrated a significant reduction in ischemia–reperfusion-related complications, such as postreperfusion syndrome, early allograft dysfunction and the incidence of symptomatic nonanastomotic biliary strictures after DCD liver transplantation¹³.

Unlike DHOPE, high metabolic activity during NMP allows for viability testing of liver grafts before transplantation. At normothermic temperatures, livers produce bile, synthesize coagulation factors and clear lactate. By analyzing these parameters, a careful assessment of liver parenchymal and biliary tree function can be performed^{14–16}. NMP has been demonstrated to reduce liver graft-related complications, especially in extended criteria donor (ECD) livers^{17–19}. However, end-ischemic NMP does not reduce rates of preservation-related complications, such as biliary strictures^{15,16,20}. In contrast to DHOPE, reintroduction of oxygen at normothermic temperatures only minimally reduces the formation of reactive oxygen species, and the recharge of depleted cellular ATP levels is complicated by the high demand for ATP in a metabolically active organ²¹. End-ischemic NMP alone is thus unlikely to sufficiently protect the liver graft against ischemia–reperfusion injury.

DHOPE and NMP serve different purposes, and combining these complementary techniques allows sequential graft reconditioning and viability assessment. The decision to perform DHOPE before NMP resulted from multiple studies reporting a beneficial effect of DHOPE before normothermic (either *in vivo* or *ex situ*) reperfusion^{8–10,22}. Although the benefit of HOPE and controlled oxygenated rewarming (COR) before NMP has not been directly compared with ‘back-to-base’ NMP alone, we believe the combination is physiologically sound and supported by experimental data^{22–24}. The combination of reconditioning using DHOPE and viability testing using NMP is valuable, especially for ECD grafts that sustain a more substantial injury during SCS. If NMP is preceded by DHOPE, the techniques should be bridged by COR

(that is, a gradual increase of perfusion temperature) to prevent cellular injury due to acute temperature shifts²⁵. This combined protocol of DHOPE–COR–NMP led to the utilization of otherwise discarded, high-risk ECD livers without an increased incidence of nonanastomotic biliary strictures^{14,26}. However, combining these complex techniques (which all require different conditions to achieve their goal) is technically challenging. This protocol provides details on how to perform this DHOPE–COR–NMP procedure in a clinical setting.

Limitations and adaptations of the protocol

The success of this protocol depends on the extent of the injury to donor organs. Although we can safely test the organs, this protocol cannot repair the damage sustained in the ischemic events before end-ischemic perfusion. Although logistically more challenging, normothermic regional perfusion (NRP) reduces the ischemic period during donor hepatectomy by restarting the abdominal circulation of the donor shortly after death with DCD donation. Therefore, NRP mitigates ischemia–reperfusion injury and reduces the incidence of biliary complications and ischemic cholangiopathy compared with rapid procurement²⁷. These results have not yet been confirmed in a randomized controlled trial. However, after NRP at the donor hospital, post-SCS ‘back-to-base’ DHOPE can still be applied at the recipient center²⁸. Alternatively, DHOPE or NMP preservation can be initiated at the procurement center to minimize the period of cold ischemia between procurement and transplantation^{8,17}. However, both techniques bring additional logistical challenges as all the equipment and manpower must be present at the donor hospital.

End-ischemic ‘back-to-base’ machine perfusion is a relatively simple approach, avoiding transportation of the perfusion device and specialized perfusion personnel at the donor hospital. Preservation by DHOPE is usually preferable to NMP as the organ is maintained in a hypometabolic state with minimal production of waste products, thereby reducing the need for adjustments to the perfusate. Obtaining experience with DHOPE is helpful before implementing the DHOPE–COR–NMP protocol.

Our protocol uses a viability testing period of 150 min NMP, which was based on previous studies suggesting adequate prediction of graft viability within this short period^{29–31}. However, other studies have used longer periods for viability testing, thereby providing additional time for a donor liver to recover and meet the acceptance criteria^{15,16}. Clinical validation data of the viability criteria and the time needed to achieve them are still emerging. Moreover, as mentioned above, our protocol was designed and optimized for the Liver Assist perfusion device (XVIVO). All critical steps and viability assessments have been tailored toward this device. Therefore, these steps and viability criteria may need to be adapted for other devices.

As an alternative to our protocol, DHOPE–COR–NMP can also be performed with a hemoglobin-based oxygen carrier (HBOC), as described previously²⁴. The major benefit of this artificial oxygen carrier to red blood cells (RBCs) is that no fluid switch is required. HBOC can be used in both hypothermic and normothermic conditions, but it is an investigational drug that has not been approved by the US Food and Drug Administration or the European Medicines Agency for human use. Owing to shortage of this investigational product and the lack of approval by competent authorities, HBOC is currently not used in our centers.

Experimental design

Donor liver procurement

For this protocol, we practice the best medical and surgical care during the retrieval of the donor liver, including short hepatectomy (preferably <45 min) and cold ischemia times (preferably below 6 h)³². Either the super rapid recovery technique or NRP should be applied for DCD liver donations³³.

Hypothermic machine perfusion

DHOPE is performed using a pressure-controlled machine perfusion device with oxygenated University of Wisconsin (UW) machine perfusion solution (1 L/min of 100% O₂). The current protocol describes dual HOPE perfusion through both the portal vein and the hepatic artery. Alternatively, single HOPE can be performed via the portal vein only (Fig. 1a). After the arrival of a donor liver in our center, we immediately start with HOPE to reduce the cold ischemic time.

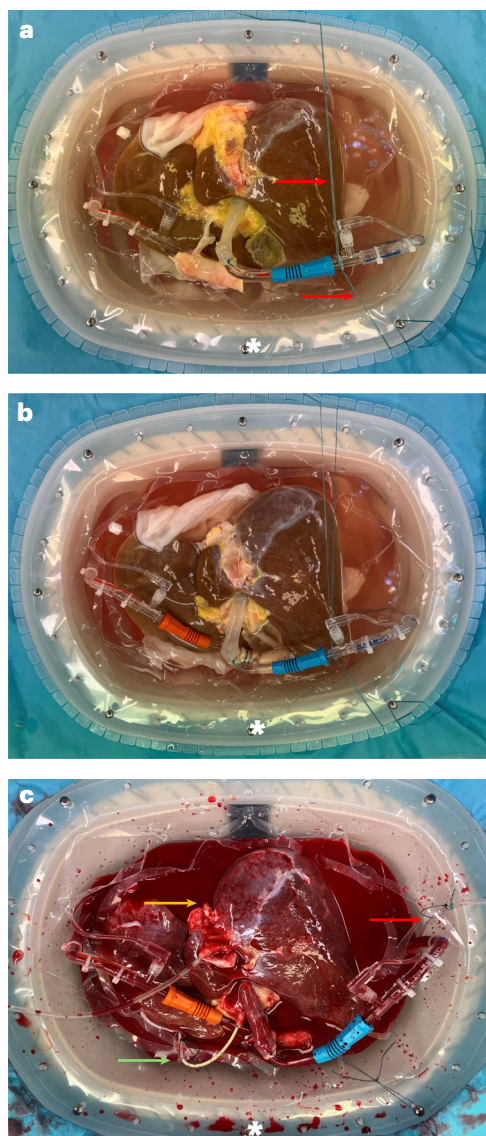


Fig. 1 | Positioning of the liver during machine perfusion. Before connection of the liver to the perfusion device, a 25 Fr cannula is placed in the portal vein. **a**, During hypothermic oxygenated machine perfusion of the portal vein, the preparation of the HA is performed, and a 25 Fr cannula is placed in the supratruncal aorta. The portal cannula can be secured with braided threads, indicated with red arrows. Hereafter, dual hypothermic oxygenated machine perfusion can be performed. **b**, Positioning of the liver during dual hypothermic machine perfusion. **c**, Positioning of the liver during normothermic machine perfusion. The red arrow indicates the medication infusion line inserted in the reservoir. The yellow arrow indicates the outlet of the venous sampling line inserted in the inferior vena cava. The green arrow points toward the position of the biliary drain. Asterisks indicate the frontal side of the Liver Assist device.

During HOPE, the regular backtable work is done to remove adjacent tissue from the donor liver and isolate the hepatic artery (HA) for cannulation and the start of DHOPE³⁴ (Fig. 1b). This enables reduction of the overall cold ischemia time by 40–45 min. The perfusion pressure in the portal vein is kept at ≤ 5 mmHg, typically resulting in a flow of 150–200 mL/min at 10 °C. The mean arterial pressure is kept at ≤ 25 mmHg, generally resulting in a flow rate of $\sim$ one-third of that in the portal vein. DHOPE is generally applied for 1–2 h^{13,35}. However, prolonged DHOPE times (up to 15 h) have recently been described to facilitate transplant logistics and daytime surgery^{36,37}. A preclinical study has shown that 1 h cold oxygenated machine perfusion is enough to switch mitochondria from a high-flux electron transfer stage to a low-flux electron transfer stage^{38,39}. Therefore, we advise performing at least 1 h of DHOPE when using our protocol. For logistical reasons, a longer DHOPE time (up to 15 h) could be considered³⁷.

Fluid change to enable COR–NMP

At perfusion temperatures below 20 °C, the solubility and diffusion capacity of oxygen in a fluid is sufficient for adequate oxygen delivery to the liver, eliminating the need for an oxygen carrier

during DHOPE⁴⁰. For NMP, the addition of an oxygen carrier to the perfusion solution is required to provide sufficient oxygen delivery at 37 °C (Table 1). In our protocol, NMP is always preceded by a short phase of 1 h of DHOPE⁴¹, after which the donor liver is disconnected from the machine and placed in a bowl with ice-cold preservation solution and melting sterile ice for a short time. The disposable tubing set of the machine perfusion device is cleansed to enable reuse during subsequent COR–NMP. After disconnection of the liver, the disposable set is quickly rinsed with sterile water and subsequently filled with the RBC-containing perfusion solution for the rewarming and NMP phases. This period of fluid change requires 15–20 min on average. Before reconnection of the liver for the rewarming phase, the organ is flushed with 1–2 L of cold Ringer’s lactate solution to remove the high amounts of potassium-containing machine perfusion solution.

Rewarming

The COR phase is started at 20 °C to prevent hemolysis of RBCs. Oxygenation is adjusted to obtain an arterial partial pressure of oxygen (pO₂) of 10–13.3 kPa. This is typically achieved with an O₂/air mixture of 21% with a flow of 100–200 mL/min. The liver is reconnected and the rewarming phase is commenced after the venous return temperature has returned to 20 °C. Subsequently, the temperature of the perfusion fluid is slowly increased to 37 °C with incremental steps of 3 °C per 7.5 min, according to previously described protocols^{14,42,43}. Pressures should be 5 mmHg and 25 mmHg for the portal venous and arterial side, respectively, at the start of rewarming. In parallel with the increase in temperature, the perfusion pressures are slowly increased using an algorithm as described in Fig. 2.

NMP

During NMP (Fig. 1c), the liver is metabolically active and hepatobiliary viability can be assessed. To evaluate viability, arterial, venous and bile samples are taken regularly and analyzed with a blood gas analyzer. If the liver meets the predefined viability criteria within the first 150 min of NMP, the liver is accepted for transplantation and NMP is continued until the recipient is ready for implantation⁴¹. The time period of 150 min is based on previous research³¹, yet longer periods of viability assessment can be used in selected cases.

Table 1 | Perfusate components for COR–NMP

Component	Manufacturer/distributor	Volume (mL)
O negative packed cells in Sag-M, hematocrit 0.6 (<2 weeks old)	Blood bank	840
Gelofusine 4%	B Braun	500
Human albumin 20%	Sanquin	250
Total parenteral nutrition (TPN N14G30E)	Hospital pharmacy	20
Addamel (trace elements)	Fresenius Kabi	10
Metronidazol (Flagyl) 5 mg/mL	Baxter BV	44
Sterile water	B Braun	335
Insulin (NovoRapid) 100 IU/mL	Novo Nordisk BV	1
Cernevit (multivitamins)	Baxter BV	2
Heparin (5,000 IU/mL)	Leo Pharma	2
Cefazolin 1 g/5 mL	Baxter BV	2
Sodium bicarbonate 8.4%	B Braun, Melsungen, Germany	55
KCl 1 mmol/mL	B Braun	2
Glutathion (Tationil) 600 mg/4 mL	Teofarma	14
Calcium gluconate 10%	B Braun	10
Total plasma volume		1,583
Total volume		2,087

KCl, potassium chloride.

Protocol

		DHOPE		COR								NMP		
Time		0:00	0:30	1:00	1:08	1:15	1:23	1:30	1:38	1:45	1:53	2:00	3:00	4:00
PV	Pressure set	3–5	3–5	3–5	3–6	4–6	5–7	5–7	6–8	6–8	6–9	6–10	6–11	6–11
	Set pressure to	3–5	3–5	3–5	4–6	5–7	5–7	6–8	6–8	6–9	6–10	6–11	6–11	6–11
	Target flow	10–20%	10–20%	20%	30%	40%	50%	60%	70%	80%	90%	100%	100%	100%
HA	Pressure set	20–25	20–25	20–25	20–25	20–30	25–35	25–35	35–45	35–45	35–55	35–55	35–60	35–60
	Set pressure to	20–25	20–25	20–25	20–30	25–35	25–35	35–45	35–45	35–55	35–55	35–60	35–60	35–60
	Target flow (mL/min)	50	50	80	100–200	100–200	100–200	200–300	200–300	200–300	300–400	300–400	400–500	500
	Temp (°C)	8–12	8–12	20	20	20	22	25	28	31	34	37	37	37
	Set temp to (°C)	Cooling	Cooling	20	20	22	25	28	31	34	37	37	37	37
	FiO ₂	100%	100%	21%	21%	21%	21%	21%	21%	21%	21%	21%	21%	21–35%
	Air flow (mL/min)	1000	1000	200	200–300	200–400	300–600	300–600	300–700	400–800	400–900	400–1000	400–1000	400–1300
	Strive pCO ₂ (kPa)	<0.73	<0.73	±10	4.5–6	4.5–6	4.5–6	4.5–6	4.5–6	4.5–6	4.5–6	4.5–6	4.5–6	4.5–6
pH	7.4													
	7.3													
	7.2													
	7.1													
Perfusion fluid		University of Wisconsin machine perfusion solution		Red blood cell based NMP perfusate										

Fig. 2 | Machine settings for DHOPE and COR–NMP. During the rewarming phase, the settings of the machine are adjusted to increase the temperature of the liver slowly. In parallel with the increase in temperature, the perfusion pressures and gas flow are slowly increased. Along with an increase in settings, an increase in CO₂ production and pH can be expected. The figure indicates a

range of settings and expected values at each time point; the actual settings depend on the liver and the results of the blood gas analyses. The red line indicates the course of the perfusate pH over time. FiO₂, fraction of inspired oxygen; PV, portal vein.

Viability testing

Testing of donor livers can be divided into two major components: the function of the liver parenchyma (hepatocellular function) and the function of the biliary tree (cholangiocellular function)^{44,45}. For hepatocellular viability testing during NMP, we use rates of lactate clearance (established by studying pH stabilization due to bicarbonate production as lactate is metabolized) and bile production as parameters^{14,15,44}. For cholangiocellular viability testing, biliary pH, bicarbonate and glucose levels are used as markers of bile duct viability^{14,31}. Notably, biliary pH, bicarbonate and glucose levels are easily influenced by corresponding perfusate values. Therefore, the difference between the biliary and perfusate pH, bicarbonate and glucose levels should be calculated, as this offers the most accurate information on biliary viability^{14,44}. Recently, the difference between bile and perfusate sodium levels was reported as a potential viability marker⁴⁶. The viability criteria used in our protocol are based on experience obtained in Groningen with combined hypothermic and NMP using the Liver Assist device (XVIVO). Therefore, these viability criteria cannot be directly extrapolated to other machine perfusion techniques or devices.

Materials

Biological materials

- Human donor liver. Human donor livers can be procured, stored, and transported to the recipient hospital as per standard practice before using this protocol
 - ▲ **CRITICAL** To avoid injury to the vascular endothelium of a vessel that may be required for anastomosis in the recipient, include a 3–4-cm-long circular segment of the suprarenal aorta with the HA to facilitate cannulation for machine perfusion. Therefore, do not make a Carrel patch of aorta at the origin of the celiac trunk. A long portal vein is preferred. The bile ducts need to be flushed with preservation fluid and the cystic duct should be identified and ligated. A cholecystectomy is not required. However, owing to edema, swelling of the gallbladder during NMP may cause compression of the portal vein, so we remove the gallbladder before the start of COR–NMP.

Protocol

Reagents

- NaCl 5% (Hospital pharmacy, cat. no. EP00367)
- O negative packed cells in Sag-M, hematocrit 0.6 (Blood bank)
- Gelofusin 4% (B Braun, cat. no. FV19513)
- Albumin 20% (Blood bank)
- Total parenteral nutrition (TPN N14G30E) (Hospital pharmacy)
- Addamel (trace elements) (Fresenius Kabi, cat. no. 016301001A)
- Metronidazol (Flagyl) 5 mg/mL (Baxter BV, cat. no. FE3400)
- Sterile water (B Braun, cat. no. 0082479E)
- Insulin (NovoRapid) 100 IU/mL (Novo Nordisk BV, cat. no. 38070065)
- Cernevit (multivitamins) (Baxter BV, cat. no. 3GM057A79)
- Heparin 5000 IU/mL (Leo Pharma, cat. no. 074227)
- Cefazolin 1 g/5 mL (Baxter BV, cat. no. 215094)
- Sodium bicarbonate 8.4% (B Braun, cat. no. 3624137)
- KCl 1 mmol/mL (B Braun, cat. no. 3616835)
- Glutathion (Tationil) 600 mg/4 mL (Teofarma, cat. no. V03AB32)
- Calcium gluconate 10% (B Braun, cat. no. 3521214)

Equipment

In addition to the materials for a regular liver transplant and liver transplant backtable procedure, the following equipment is needed:

Sterile consumables

- Liver Assist disposable (XVIVO, cat. no. ME H13 3411)
- Portal vein cannula (25 Fr, curved; XVIVO, cat. no. 11.01.520)
- HA cannula (25 Fr, straight; XVIVO, cat. no. 11.01.519)
- Three-way valves (with extension tube; Codan, cat. no. 71.4556)
- Nutrifit 8 Fr feeding tube (Vygon, cat. no. 1362.087)
- Extension lines 150 cm (Codan, cat. no. 71.4021)
- Gauzes 10 × 10 (BSN Medical, cat. no. 72227-01)
- Gauzes 45 × 45 (Medeco BV, cat. no. 115 062)
- Syringes 20 mL (Luerlock, BD, cat. no. 300629)
- Syringes 10 mL (B Braun, cat. no. 4616103V)
- Syringes 60 mL catheter tip (BD, cat. no. 300867)
- Syringes 60 mL luerlock (BD, cat. no. 300865)
- Infusion line (V-green extension, Vygon, cat. no. 71100.20)
- Blood gas syringe (Radiometer Medical ApS, cat. no. 956-622)
- Connecting suction tubing (Extrudan Surgery ApS, cat. no. 2302 3500 10 GR)
- Suction jar 3 L (BactiClear Antimicrobial Suction Liner; VacSax, cat. no. VAL-301)
- Infusion system (Dispo Medical, cat. no. 18125)
- Sterile transparent cover for scale (Angiokard, cat. no. 1900670)
- Sterile bowls for liver (Angiokard, cat. no. 1900387)
- Mersilene 1.0 75 cm (Ethicon, cat. no. EH6936H)
- Cold storage bags liver (Beldico SA/NV, cat. no. HPF6397)
- Bag-a-jets (ECM Europe, cat. no. A-2000)

Storage/flush

- UW machine perfusion solution (CarnaMedica, cat. no. PPRT)
- Cold storage preservation solution (Bridge to Life, cat. no. BTLBUW-1000)
- Crushed ice sterile (frozen saline; Baxter, cat. no. 287231K)
- Sterile water (1,000 mL; Fresenius, cat. no. B230531)
- Cold Ringers lactate (1,000 mL; Baxter, cat. no. WKE2324)
- Crushed ice, nonsterile for cooling the water bath
- Demineralized water for filling the thermo unit of the Liver Assist
- Small buckets of tap water for tricking the flow sensors

Hardware

- Gas mixer (compressed air + O₂)
- Compressed air supply
- 100% Oxygen supply
- Gas flow controller 0–2 L
- Surgical vacuum system
- Infusion pump
- Tubing clamps
- Blood gas analyzer (ABL90Flex blood gas analyzer, Radiometer) that measures lactate (mmol/L), glucose (mmol/L), sodium (cNa⁺), potassium (cK⁺), chloride (cCl[−]), calcium (cCa²⁺), pH, pCO₂, pO₂, oxygen saturation (sO₂), bicarbonate (cHCO₃[−](p)c) and arterial base excess (BEC)
 - ▲ **CAUTION** Ensure the limits of the measuring range of the blood gas analyzer are set to their maximum.
- Weight scale (0–10 kg)
- Liver Assist device (XVIVO)

Reagent setup

NMP perfusate

● TIMING 30 min

Prepare the perfusion fluid by adding the reagents listed in Table 1 together.

▲ **CAUTION** Pharmaceuticals should be prepared by the hospital pharmacy according to good manufacturing practice. Do not merge the pharmaceuticals with the other components (RBC, gelofusin and albumin) until infusion into the machine perfusion device. Only add calcium gluconate just before starting NMP to avoid the formation of undesired crystals.

■ **PAUSE POINT** The combined pharmaceuticals as prepared by the hospital pharmacy (without calcium gluconate, RBC, gelofusin and albumin) can be stored at 4 °C for up to 1 week.

Supplementations

Gather the following medications for potential supplementation during NMP:

- Calcium gluconate 10% (±30 mL)
- KCl 1 mmol/mL (±30 mL)
- Sodium bicarbonate 8.4% (±100 mL)
- Heparin 5,000 IU/mL (5 mL)
- NaCl 5% (±50 mL)
- Glucose 50% (±20 mL)

Equipment setup

Currently, the only commercially available machine perfusion devices that allow both dual, oxygenated hypothermic and NMP are the Liver Assist device (XVIVO) and the PerLife device (Aferetica). We have developed the DHOPE–COR–NMP protocol using the Liver Assist. The Liver Assist is a pressure-controlled dual perfusion machine that generates a pulsatile flow through the HA and a continuous flow through the portal vein. It is an open perfusion system in which the HA and portal vein are cannulated, and the preservation solution flows passively from the vena cava into the reservoir. Follow the steps below for perfusion device setup.

▲ **CRITICAL** To avoid contamination of the perfusate and the donor liver, ensure that the inside of the circuit remains sterile during all steps.

Perfusion device setup

● TIMING 10 min

- Prepare the perfusion device according to the instructions of the manufacturer
- Connect the filling lines and sample ports provided with the Liver Assist disposable set to the arterial and portal circuit

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- Remove the white cap from the short line on the portal oxygenator and attach a three-way valve. This is needed later in the procedure for connection of the heparin infusion pump during NMP (Fig. 3a)
- Connect both oxygen tubes of the oxygenators to the Y-connector provided with the disposable set
- Connect the gas mixer outlet to the Y-connector on the oxygen tubing and set the 100% oxygen flow rate to 1 L/min for oxygenation during DHOPE

Procedure

According to hospital practice, the potential recipient is present in the hospital during the DHOPE–COR–NMP protocol. This way, the recipient surgery can be commenced immediately once the liver meets the viability criteria. The potential recipient may be sent home if the viability criteria are not met. A video demonstrating the essential steps of the protocol is available as Supplementary Video 1, providing a visual guide to the procedure described below. Moreover, our centers have started using an International Training Curriculum for Organ Perfusionists, which includes hands-on training⁴⁷. An overview of all personnel involved in the procedure is summarized in Supplementary Table 2.

Priming the circuit

● TIMING 20 min

1. Fill the circuit with 3 L of sterile UW machine perfusion solution through the filling lines.
2. Prime the machine according to the manufacturer's instructions. Do not detach the filling lines, as you need them for refilling the disposable set later in the procedure. Instead, disconnect the bags with UW perfusion solution once they are empty to clear both filling lines completely and put the sterile cap back on to keep the tip sterile for later use. Connect a three-way valve to the portal oxygenator to allow portal vein perfusate sampling (Fig. 3a).
3. Set the arterial pressure to 25 mmHg and the portal pressure to 5 mmHg. Set the temperature to 'Cooling'. Check to make sure the red rotors in the tubing of the thermo unit are spinning.
4. Take a blood gas measurement to check the oxygenation of the perfusate (>100 kPa).
▲ **CRITICAL STEP** Discard the first 5 mL taken from the sample port line before taking your sample to make sure the sample provides a real-time measurement
5. Put nonsterile crushed ice in the thermo unit to reach the desired temperature of 8–12 °C.
■ **PAUSE POINT** The primed circuit can wait for a few hours until the liver is ready for connection.

Backtable procedure

● TIMING 30 min

Our backtable procedure is comparable to a regular backtable procedure for a donor liver before transplantation, but now the portal vein and HA are cannulated. As cold ischemia time is a major risk factor for the development of biliary injury, part of the backtable procedure is performed during single HOPE perfusion³⁴.

6. Remove the remaining parts of the diaphragm and dissect the vena cava and the portal vein. Place the (25 Fr) cannula in the portal vein and make sure the entrance of the cannula points toward the right liver lobe. An essential aspect of cannulation of the portal vein is preventing any twists in the portal vein main stem and the left and right branches. Secure the cannula with a braided thread (Fig. 1a).
▲ **CRITICAL STEP** In the case of a large gall bladder, a cholecystectomy is sometimes required to allow adequate positioning of the portal vein cannula during perfusion. As an alternative, the gallbladder can be emptied using a syringe and large bore needle.

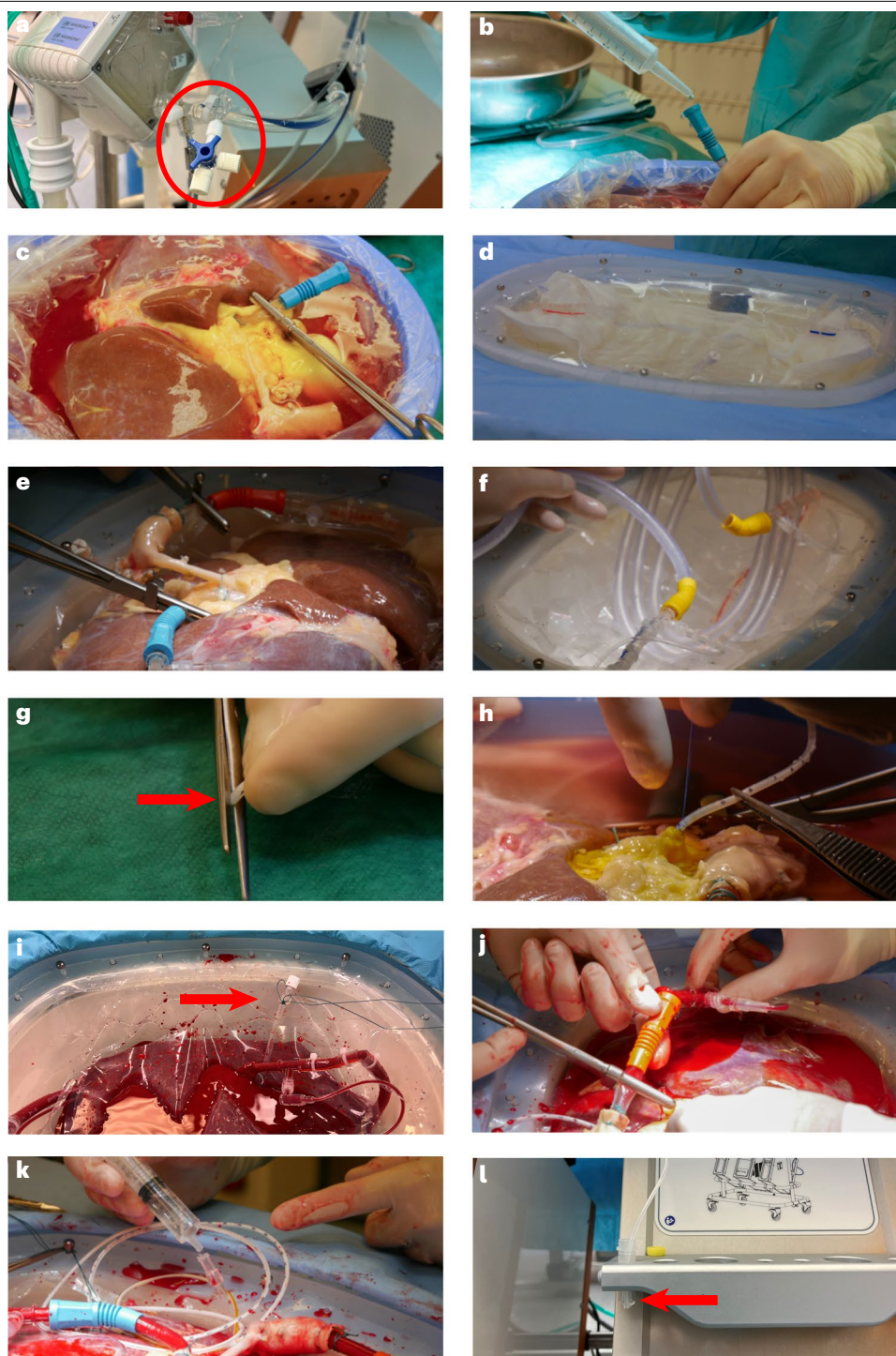


Fig. 3 | Essential steps of the DHOPE-COR-NMP procedure. **a**, Connect a three-way valve (red circle) to the portal oxygenator (Step 2). **b,c**, Fill the cannula with a syringe to check for leakages before flushing (**b**) and clamping the filled cannula (Step 7) (**c**). **d**, Place a large gauze in the reservoir before initiating HOPE (Step 8). **e**, Placing hose clamps on the cannulas before stopping DHOPE (Box 1, step 4). **f**, Connect the suction tubes to the perfusion set (Box 1, step 9). **g**, Adjust the tip of the bile drain (red arrow) to ensure bile flow (Step 22). **h**, Secure the bile drain

in the extrahepatic bile duct with a purse string suture and an additional braided suture around the cannula (Step 22). **i**, Secure the bile drain to the reservoir with a braided thread (red arrow) (Step 22). **j**, Reconnect the donor liver to the circuit using the clamps (Box 1, step 26). **k**, Flush the bile drain with saline or water to ensure a lower resistance while positioning the bile drain (Box 1, step 27). **l**, Place a drop of mineral oil in the Eppendorf cup (red arrow) with the tip of the bile drain cannula positioned under the mineral oil level (Step 30).

7. Flush the portal vein with at least 1 L of UW machine perfusion solution to replace the cold storage preservation solution within the liver. Check for leakages during the back table flush and close any remaining holes and side branches of the portal vein. If the effluent is not clear yet, flush another bag of UW machine perfusion solution through the portal vein. At the end of the flush, close the cannula with a large hose clamp to stop the flush and leave the portal vein filled with preservation solution. This makes it easy to connect the liver to the machine without air bubbles.
▲ CRITICAL STEP Prevention of air bubbles in the cannula and portal vein is very important. An air bubble can cause an airlock and reduce flow to zero, especially during DHOPE, when low pressures are used. Air bubbles can be hard to detect as perfusion solutions for DHOPE are colorless. We advise using a large 60 mL catheter tip syringe to first aspirate the portal vein and remove potential air bubbles, and then fill the cannula completely and place a hose clamp before connecting the flush (Fig. 3b,c).

HOPE phase

● TIMING 30 min

- ▲ CRITICAL** Ensure the perfusate temperature is between 8 °C and 12 °C before connecting the liver to the perfusion device.
8. Place an unfolded large gauze (40 × 40 cm) on top of the net in the reservoir of the Liver Assist disposable set to prevent debris from entering the circuit that may be released during dissection of the HA (Fig. 3d).
9. Position the liver with the dorsal side upward and the gall bladder positioned toward the front of the machine (Fig. 1a).
10. Connect the portal cannula to the portal inlet tube and press 'OK' at the portal pump unit to start portal perfusion. Check for air bubbles and remove the hose clamp from the cannula to start the perfusion. Aim for a perfusate flow between 100 and 300 mL/min with a pressure between 3 and 5 mmHg. The highest possible flow at the lowest possible pressure is desired.
▲ CRITICAL STEP After connection, you will receive a high-pressure alarm as the hose clamp is still on the cannula. Carefully check for air bubbles, remove the hose clamp, disengage the alarm and reset the pressure to the right setting (3–5 mmHg).
- ◆ TROUBLESHOOTING**
11. Fix the position of the portal cannula (Fig. 1a). If a very long portal vein is present, make sure not to pull the portal vein too strongly toward the caudal side of the liver. To prevent this, a braided thread can be secured to both the caudal and cranial sides of the liver (red arrows).
12. Proceed with preparation of the HA and the coeliac trunk. Dissect the pancreatic head and perform an arterial reconstruction if necessary. Close the distal end of the aorta segment with a running prolene 2-0 suture.
▲ CRITICAL STEP When manipulating the hepatic artery, the portal flow can change. Keep an eye on the portal flow and make sure there is a sufficient flow.
13. Place the arterial cannula in a circular segment of the suprarenal aorta (Fig. 1b). Make sure all side branches are properly closed and the oversewn infrarenal aorta segment does not leak. Use a 60 cc syringe with a catheter tip to check for leakages. Carefully close the HA in the hilus by hand and fill the cannula with UW machine perfusion solution. By manually applying a slight pressure in the artery, potential leakages can be detected. When satisfied, the cannula can be closed with a hose clamp once filled to facilitate an air bubble-free connection. Alternatively, a side branch, such as the gastroduodenal artery, can be left open to purge air after starting perfusion. These can then be tied off or clamped after starting perfusion.
▲ CRITICAL STEP When the circular aorta segment is absent or unusable, direct cannulation of the celiac trunk is an alternative option. The manufacturer recommends using alternative cannulas (Hagmed, cat. no. 05.01.504) available in sizes 8/10/12 Fr. The small diameter of these cannulas causes a substantial loss of pressure. This must be corrected manually in the pressure setting, following the pressure loss graphs provided in the instructions for use of the device. The combination of the cannulas with the smallest diameter (8/10 Fr) and the high arterial flows achieved during NMP (300–500 mmHg) causes a pressure loss that cannot be corrected in the machine pressure setting owing to a software limitation. This is

not a problem for DHOPE owing to the low flows; however, for NMP, it is only possible to use the 12 Fr cannula if no circular aorta segment is usable. If placement of a 12 Fr cannula is not possible, the best alternative option would be to first suture a small segment of donor iliac artery to the celiac trunk or common HA and use this iliac artery conduit for placement of the 25 Fr cannula.

DHOPE phase

● TIMING 1 h

▲ **CRITICAL** The DHOPE phase can be extended if this is more convenient owing to logistical considerations^{36,37}. Shortening is not recommended, as the goal of the DHOPE phase will then not be achieved^{38,39}.

14. Connect the arterial cannula to the arterial inlet tube and press 'OK' at the arterial pump unit to start arterial perfusion. Aim for a perfusate flow of one-third of the portal flow with a pressure between 18 and 25 mmHg. The arterial flow will most probably increase during the perfusion. Reduce the pressure if necessary to prevent hyperperfusion

◆ TROUBLESHOOTING

15. Place a small, wet gauze to cover the vessels to prevent dehydration.
16. Place the lid on the reservoir for isolation and sterility. Pay attention to the positioning of the cannulas to ensure a stable flow after placement of the lid.
17. Take a blood gas reading every 30 min to check the oxygenation of the perfusate (>100 kPa). Discard the first 5 mL taken from the sample port line before taking your sample.
18. Prepare the fluid switch by gathering the following:
 - Two small buckets of water (to place the flow sensors in during the fluid switch)
 - The components for the perfusate; three units of packed RBCs, the infusion bag with medication, gelofusin, and albumin. Allow these to reach room temperature (18–20 °C)
 - 3 L bottles of sterile water for rinsing the disposable set
 - Vacuum system
 - Bag-a-jets
 - A sterile bowl with ice and preservation solution to store the liver
 - 2 L bags of cold Ringer's lactate for flushing the liver. Keep these on ice until use
 - Nutrifit feeding tube 8 Fr for cannulation of the bile duct
 - Two extension tubes (150 cm) and two three-way valves for the inferior vena cava drain and medication line
19. Prepare a 50 mL infusion syringe with 45 mL NaCl 0.9% and 5 mL heparin 25,000 IU/5 mL.

▲ **CRITICAL STEP** To reduce cold ischemic time, ensure everything is ready for the fluid switch before the liver is disconnected.

Fluid switch

● TIMING 30 min

20. Follow the steps in Box 1 for the perfusate switch and the start of the rewarming phase.
21. Cut the closed distal end of the bile drain (pediatric feeding tube, 8 Fr) to ensure bile can flow freely through both the tip and one side hole (Fig. 3g).
22. Place a bile drain in the distal end of the common bile duct after performing a purse string suture with a 5-0 monofilament suture (Fig. 3h). Tie an additional 4-0 braided suture around the cannula in the distal end of bile duct.

▲ **CAUTION** Make sure to excise and discard the distal end of the common bile duct before bile duct anastomosis in the recipient.
23. Place a catheter in the inferior vena cava and secure it with a suture in the vena cava wall (Fig. 1c). The tip of the drain is placed in the cranial opening of the inferior vena cava (yellow arrow). Make sure the catheter still allows aspiration after the sterile lid has been placed onto the reservoir.

▲ **CRITICAL STEP** Measure the weight of the donor liver to calculate a strive portal venous flow during NMP. Recent research, albeit in vivo, demonstrated an optimal portal venous flow of 92–110 mL/min/100 g liver weight^{48,49}. Strive flow for HA is approximately one-third of the portal vein flow. For the DHOPE phase, no optimal flows have been reported.

BOX 1

Fluid switch

● TIMING 30 min

1. Connect the bag with gelofusin to the portal filling line. Do not open the white clips yet.
2. Connect the bag with albumin to the arterial filling line. Do not open the white clips yet.
3. Stop perfusion by pausing the pump head of the portal and arterial pump units.
▲ CRITICAL STEP Do not switch off the pump units yet.
4. Place a hose clamp on the portal and arterial cannula and disconnect the liver from the device (Fig. 3e). Store it in a bowl with cold preservation solution and sterile ice. Start performing Steps 21–23 of the main procedure at the backtable simultaneously with steps 5–23 of Box 1.
5. Immediately turn down the oxygen flow to 21% O₂ at 150 mL/min.
6. Immediately set the temperature of the thermo unit to 20 °C.
7. Remove the gauze from the reservoir.
8. Disconnect both flow sensors from the tubes and place them each in a bucket of water, to prevent device errors during priming.
9. Empty the reservoir with the vacuum system. Place the suction tubes sterile in the reservoir.
10. As soon as the reservoir is empty, connect the suction tubes (sterile) to the arterial and portal inlet tubes (Fig. 3f).
11. Disconnect both pump heads from the pump units and keep them horizontal for proper and rapid emptying.
12. Once the circuit is empty, fill the reservoir with 3 L of sterile water to rinse the circuit and wash away all the remaining UW.
13. Disconnect the vacuum tubing (sterile) as soon as the rinsed circuit is empty.
14. Reconnect both pump heads.
15. Open the white clips of the filling lines and start filling the circuit with the gelofusin and albumin.
▲ CRITICAL STEP Only start filling the system once the vacuum system is disconnected.
16. Bag-a-jet the three units of RBCs and the infusion bag with medication into the reservoir.
17. Replace the filling lines with a sterile cap when the infusion lines have been emptied.
18. Restart both pump units.
19. Repeat the priming steps as described by the manufacturer.
▲ CRITICAL STEP Do not forget to de-air the pressure lines again before pressure zeroing.
20. Set the portal pressure to 5 mmHg, the arterial pressure to 25 mmHg and the temperature to 20 °C.
21. Reconnect the flow sensors to the tubes.
22. Take a blood gas reading to check all parameters and make sure the values are within the expected range (Supplementary Information 3). Do not correct the pH yet, as pH will be low owing to high CO₂ levels. This CO₂ will disappear rapidly without a liver connected to the perfusion device.
23. Secure the tip of the bile tube in the reservoir above perfusate surface level with a braided thread (Fig. 3i).
24. Perform a 2 L Ringer's Lactate flush at the backtable: 1.5 L for the portal vein and 0.5 L for the hepatic artery.
25. Cover the sterile field of the Liver Assist device with the new sterile drape provided with the disposable set.
26. Reconnect the liver to the Liver Assist device as described in Steps 9–11 and 14 of the main procedure and start rewarming (Figs. 1c and 3j).
27. Position the bile drain through the tube in the reservoir and place the tip in an Eppendorf cup to collect the bile.
▲ CRITICAL STEP Flush the tube with some saline first to reduce resistance (Fig. 3k).
28. Position the end of the venous sample line outside the sterile field.
29. Position a 150 cm extension line from the reservoir outside the sterile field to facilitate the administration of medication during COR–NMP (Fig. 1c, red arrow).
30. Place a small wet gauze on the vessels to prevent dehydration.
31. Place the lid on the reservoir without kinking the venous sample line and medication line.

Rewarming phase

● TIMING 1 h

24. Follow the rewarming settings as presented in Fig. 2.

◆ TROUBLESHOOTING

25. Take an arterial blood gas reading every 7.5 min to check parameters. Correct for electrolyte disturbances and adjust gas flow if necessary. Follow the supplementation scheme in Table 2.
▲ CRITICAL STEP The recommended ratio of gas flow to blood flow for the oxygenators in the circuit is 1:1. To keep the CO₂ buffer capacity stable, the ratio used in this protocol is far below 1:1, initiating a risk of a ventilation/perfusion mismatch at high portal flows. For this reason, saturation of the perfusate in the portal circuit after the oxygenator (60–80%) is much lower compared with the arterial circuit, approaching the *in vivo* physiological saturation of portal blood. Closely monitor the saturation in the vena cava (55–75%) and the saturation of the arterial (95–100%) inflow to avoid hypoxia. Pay attention to the color of the portal and arterial venous blood throughout the perfusion, as a darkening color is the

Table 2 | Adjustments during normothermic machine perfusion

Blood gas	Measurement	Strive value	Physiological context	Medication	Comments for correction
Arterial	Metabolites	Lactate (mmol/L)	<1.8 mmol/L At 150 min NMP	Lactate is high at the start of the procedure and will continue to rise during rewarming. A properly functioning liver clears lactate. Rising lactate after rewarming is therefore a sign of liver dysfunction or hypoxia	N/A
		Glucose (mmol/L)	>8.0 mmol/L	Decreases when there is sufficient aerobic metabolism	Glucose 50% 2,775 mmol/mL
Electrolytes	Sodium (cNa ⁺)	135–145 mmol/L	At 20°, sodium is low, but this corrects itself automatically ~33–34 °C when the Na/K-ATPase pumps start to function	NaCl 5% 0.86 mmol/mL	Correction for hyponatremia is rarely necessary. Hyponatremia cannot be corrected. Beware of the Na ⁺ in NaHCO ₃
	Potassium (cK ⁺)	3.5–5.0 mmol/L	High at the start of rewarming. Only drops when metabolism starts and the Na/K-ATPase pumps begin to function. Drops quickly during rewarming, so be prepared to administer in time	KCl 74.6 mg/mL 1 mmol/mL	Correct for hypokalemia. 2–8 mL per bolus
	Chloride (cCl ⁻)	>90 mmol/L	Cl is exchanged for HCO ₃ ⁻ via the AE2 transport protein: alkaline bile is therefore dependent on Cl ⁻	NaCl 5% 0.86 mmol/mL	10–20 mL per bolus. Only correct during NMP (not during rewarming) and keep an eye on the sodium levels during administration
	Calcium (cCa ²⁺)	≥0.70 mmol/L	Given the role of calcium in the development of ischemia–reperfusion injury, we advise not to supplement before the start of NMP	Calcium–gluconate 0.225 mmol/mL	Normally 10–20 mL is needed after rewarming
Oximetry	pH	7.35–7.45	At the end of rewarming, there is always a physiological pH drop due to the high lactate levels. Do not correct the pH too aggressively or too early, since a mild acidic environment during rewarming protects against ischemia reperfusion injury. Keep an eye on your base excess for differentiating between respiratory/metabolic influence pH. Start correcting when BEC <–10 (see HCO ₃ ⁻ and BEC)	NaHCO ₃ 8.4% 1 mmol/mL or compensate respiratory (see pCO ₂)	5–10 mL per bolus. Start correcting when BEC <–10 during rewarming
	pCO ₂	4.5–6.0 kPa	Acidotic end product of metabolic activity and a substrate of HCO ₃ ⁻ . You can reduce excess CO ₂ by increasing oxygen flow (hyperventilation) and build up more CO ₂ by decreasing oxygen flow (hypoventilation). During rewarming, there is still little metabolic activity and thus little CO ₂ production: be careful with the ventilation	↓gas flow = ↓pH ↑gas flow = ↑pH	Consider an increase in gas flow in an acidic environment and pCO ₂ > 5.5 kPa
	pO ₂	10–13 kPa	Hyperoxia is toxic because of the formation of oxygen radicals. Slow deterioration of oxygenators can cause a stable pO ₂ to drop unexpectedly during longer NMP durations. Keep a close eye on this and correct if necessary	FiO ₂ (%)	Adjust in steps of 1–2%
	sO ₂	≥93%	Make sure the arterial saturation is above 93% to avoid hypoxia	FiO ₂ (%)	No indication. Be guided by venous sO ₂ and arterial pO ₂
	Bicarbonate (HCO ₃ ⁻)	>20 mmol/L	Alkalotic substrate for pH correction that decays into H ₂ O and CO ₂ . Physiological increase during NMP due to lactate clearance. Be aware that HCO ₃ ⁻ decreases if CO ₂ levels are <4.5 mmol/L	NaHCO ₃ 8.4% 1 mmol/mL	If base excess <–10.0, be careful if the liver still has to clear lactate, as this results in a physiological increase in HCO ₃ ⁻
	Base excess (BEC)	+3.0 to –3.0	The metabolic (non-respiratory) component of the acid/base balance. BEC is metabolic, and therefore cannot be corrected by changing the gas flow	NaHCO ₃ 8.4% 1 mmol/mL	Consider 10 mL of sodium bicarbonate if BEC <–13 during rewarming and start NMP

Table 2 (continued) | Adjustments during normothermic machine perfusion

Blood gas		Measurement	Strive value	Physiological context	Medication	Comments for correction
Venous gas	Oximetry	FiO ₂	55–75%	Percentage of hemoglobin saturated with O ₂ . It should remain above 55%	FiO ₂ (%)	Correct venous FiO ₂ (55–75%) and arterial pO ₂ (10–13 kPa) by changing the FiO ₂ setting
Bile	Metabolites	Glucose	N/A	Glucose is normally reabsorbed by cholangiocytes. A high glucose concentration indicates a poor bile duct viability. Absolute value is irrelevant, always compare with the arterial glucose	N/A	N/A
	Blood gas	pH	>~7,45 (arterial pH)	Alkaline bile protects the bile ducts against caustic bile salts. An insufficient rise in pH during NMP is a sign of poor bile duct viability. Bile pH is dependent on Cl ⁻ and arterial pH	N/A	N/A
		Bicarbonate (HCO ₃ ⁻)	~18 mmol/L (arterial HCO ₃ ⁻)	Absolute value is irrelevant, always compare with the arterial HCO ₃ ⁻	N/A	N/A
	Oximetry	Bilirubin	N/A	One hepatocyte function is to excrete bilirubin into bile. Important to realize that this is a hepatocyte function and not a cholangiocyte function	N/A	N/A

N/A, not applicable.

first sign of inadequate oxygenation. We do not routinely take blood gases from the portal circuit but rely on blood gases taken directly from the vena cava in setting the FiO₂.

◆ TROUBLESHOOTING

NMP phase

● TIMING 2.5 h

26. Connect the syringe with diluted heparin with an infusion line to the three-way valve on the portal oxygenator (Fig. 3a). Start the heparin infusion pump at a rate of 2 mL/h, up to a total of 1,000 IU heparin/h.
27. Handle machine settings as described in Fig. 2.
28. Take an arterial blood gas reading every 30 min to check parameters. Correct electrolytes and adjust the gas flow based on the supplementation scheme in Table 2.
29. Take a venous blood gas reading every 30 min to check oximetry values. Adjust the FiO₂ setting if necessary. Follow the supplementation scheme in Table 2.
30. Take a bile gas reading every 30 min to assess bile duct viability.
Empty or replace the bile collection tube before every gas analysis of the bile to ensure an accurate measurement of the bile composition. Put a few drops of mineral oil in the tube to cover the bile surface and prevent gas exchange between the bile and the air (Fig. 3l, red arrow). Then place the bile drain at the bottom of the cup underneath the mineral oil.
▲ **CRITICAL STEP** A blood gas analyzer is not designed to measure bile composition. Although we use the values measured by the blood gas analyzer to assess the viability of the liver, the actual values might slightly differ. Certain built-in corrections based on blood composition will not apply to bile, and some values will be outside the measuring range.
◆ **TROUBLESHOOTING**
31. Decide whether to proceed with the transplant based on the viability criteria defined in Table 3. End the transplant procedure if the liver does not meet the defined viability criteria (that is, the risk of post-transplant complications is too high to continue the transplant).

Continuation of NMP until transplant

● TIMING 3–5 h

32. Start the recipient surgery.
33. Continue perfusion of the donor liver until implantation.
▲ **CRITICAL STEP** Keep track of the arterial blood gasses every 30 min. Glucose may need to be administered at some point, as well as an increase of the FiO₂ or airflow. Follow the supplementation scheme in Table 2.

Table 3 | Hepato-biliary viability criteria that must be reached within the first 150 min of NMP to proceed with transplant

	Parameter	Green zone (official viability criteria)	Orange zone (borderline values that could be acceptable if the other viability criteria are 'green')	Red zone (do not meet the viability criteria)
Hepatocytes	Bile production (g)	$\geq 10^a$	5–10	<5
	Perfusate lactate (mmol/L)	<1.7	1.7 to 4.0	>4.0
	Perfusate pH	7.35–7.45	7.25 to 7.35	<7.25
Cholangiocytes	Bile pH	>7.45	7.40 to 7.45	<7.40
	Δ pH	>0.10	0.05 to 0.10	<0.05
	ΔHCO_3^- (mmol/L)	>5.0	3.0 to 5.0	<3.0
	Δ Glucose (mmol/L)	<–5.0	–5.0 to –3.0	>–3.0

^aOf which ≥ 4 g in the last hour. Δ indicates the bile value minus the perfusate value. Adapted with permission from ref. 26, Elsevier.

34. At the end of machine perfusion, take a sample of the NMP perfusate for culture to check for any microbiological contamination of the perfusate.
35. Upon cessation of machine perfusion, flush the liver with at least 2 L of ice-cold preservation solution, and place it in a bowl with preservation solution with sterile ice.
36. Transplant the donor liver into the patient.

Troubleshooting

Troubleshooting advice can be found in Table 4.

Table 4 | Troubleshooting table

Step	Problem	Possible reason	Solution
10, 14	Flow too low	Obstruction of vena cava by gauze or net	Remove the obstruction and make sure an outflow is possible
		Kinking vessel	Check if there is a kinking vessel that obstructs the flow through the liver. Reposition the cannula(s) with a braided thread
		Collapsed portal vein due to the positioning of the cannula	Try to reposition the cannula with a braided thread toward the cranial side of the liver
		Air bubbles in portal vein or HA cannula can cause an air lock and obstruct the flow	Reconnect the cannula without air in the system. When reconnecting the liver, you must manually maintain a flow of ± 250 mL/min by increasing/decreasing the pressure setting. Immediately turn back the pressure to normal as soon as the cannula is reconnected
		Liver at a higher position than pressure sensor	Make sure the reservoir is properly unfolded. If the cannula is still at a higher position than the pressure sensor, cut the net to lower the liver in the reservoir. <i>Note:</i> Do not cut the net too much. The perfusate volume is lower during NMP and the liver might block the venous inlet. If the liver is large (for example, >2 kg), reduction of the circulating volume by 500 mL can improve flow during DHOPE by lowering the position of the liver in the reservoir
		High resistance in vascular bed	Cut the net to manually increase the pressure to 0.5 mmHg. This is particularly likely to be needed during rewarming and NMP
		Software fault/air in pressure sensor	Try to restart the pump unit. Perhaps the pressure sensor has not been zeroed properly
	Flow too high	Leakages	This is the most likely reason for non-physiological high flows. Check if there is venous outflow and check for leaking side branches and dissections. Clip or stitch all side branches to make sure there is a sufficient flow through the liver
		Incorrect pressure measurement (after ruling out leakages)	Check if the valves of the pressure sensors are open and the arterial and portal pressure sensors are not mixed up
		Low resistance in vascular bed (after ruling out leakages and incorrect pressure measurement)	If there are no leakages and the pressure is being measured correctly, there is probably a low resistance in the vascular bed. Perfusion can be continued at low pressure

Table 4 (continued) | Troubleshooting table

Step	Problem	Possible reason	Solution
24	The set temperature cannot be reached or the temperature is out of range	The liver is still too cold and is cooling down the perfusate	It can take a while for the liver to warm up, especially if there is a high vascular resistance. To avoid endothelial damage, slowly increase the temperature, patiently wait for the liver to warm up, and do not increase the pressure. You will catch up with the settings in Table 2 later during rewarming. The resistance will decrease once the temperature is higher, and so the temperature will further increase
25	Saturation of portal inflow is too low/venous saturation is too low	Ventilation/perfusion mismatch in portal oxygenator	Increase the FiO ₂ . Try to avoid blowing off all CO ₂ by turning up the gas flow, since this will result in buffer capacity issues
	pH <7.30 but the lactate has already been cleared from the perfusate	Hyperventilation	Owing to a high gas flow setting, the CO ₂ concentration is below 4.5 kPa, resulting in a respiratory acidosis. First correct the oxygenation settings, and then look at the sodium concentration to see if you still can supplement the bicarbonate with sodium bicarbonate
		Metabolic acidosis	The liver is unable to neutralize the environment. This can be caused, for example, by an accumulation of ammonia. Consider correcting this metabolic acidosis with a small bolus of sodium bicarbonate 8.4%. Make sure that the sodium concentration remains within limits
30	No or low bile production	Lumen of bile drain vacuum too close to wall of extrahepatic bile duct	Reposition the bile drain slightly and monitor the bile flow from the tip of the drain
		Bile duct twisted	Reposition the bile duct and untwist
		Bile drain obstructed by mersilene used as ligature	Remove the obstructing mersilene and resecure the drain by a new less tight ligature
		Poor liver function	In our experience, even livers with severe graft dysfunction still produce bile. Therefore, always reconsider the abovementioned technical errors first before assuming poor liver function as the cause of no bile production
	Bloody bile production	Leakage of perfusate into bile drain	Secure the bile drain with an extra ligature that is tighter than the current one

Timing

Figure 4 shows a timeline of the procedure. To minimize the cold ischemic time, we recommend preparing for the procedure before the liver has arrived. All set-up steps such as the preparation of the NMP perfusate and setting up the perfusion device can be performed in advance.

Preparations

Reagent setup: 30 min

Equipment setup: 10 min

Procedure

Priming circuit (Steps 1–5): 20 min

Backtable procedure (Steps 6–7): 1 h

HOPE (Steps 8–13): 30 min

DHOPE (Steps 14–19): 1 h (this part can be prolonged if it benefits logistics)

Fluid switch (Steps 20–23): 30 min

Rewarming phase (Steps 24–25): 1 h

NMP viability assessment (Steps 26–31): 2.5 h

Continuation of NMP until transplant (Steps 32–36): 4–6 h

Anticipated results

This protocol of combined DHOPE and NMP allows for resuscitation and viability testing of donor livers before transplantation, increasing the absolute number of potential donor livers by improving the utilization rate. So far, more than 250 donor livers have been perfused with the DHOPE–COR–NMP protocol at our two centers. Of these donor livers, all but two met the

Protocol

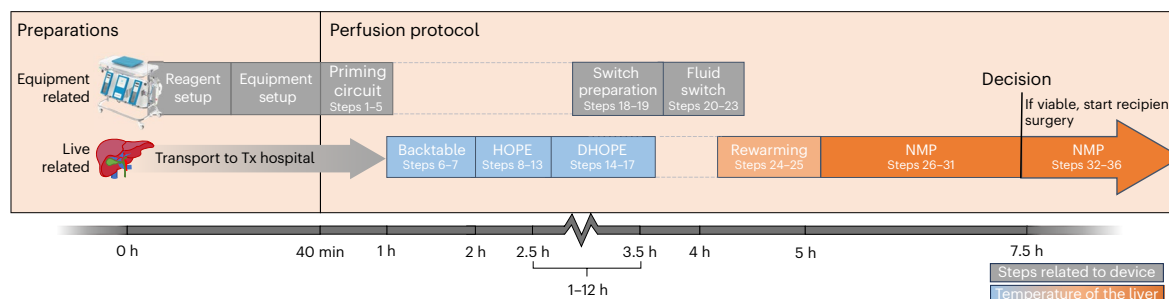


Fig. 4 | Protocol workflow. Timeline indicating the required time for each step in the protocol.

hepatocellular viability criteria (Fig. 5a,b), and 64% also met the biliary viability criteria (Fig. 5c-f). In our experience, the utilization rate experienced with this protocol is therefore 64%. However, the utilization rate strongly depends on the type of high-risk donor livers accepted for inclusion

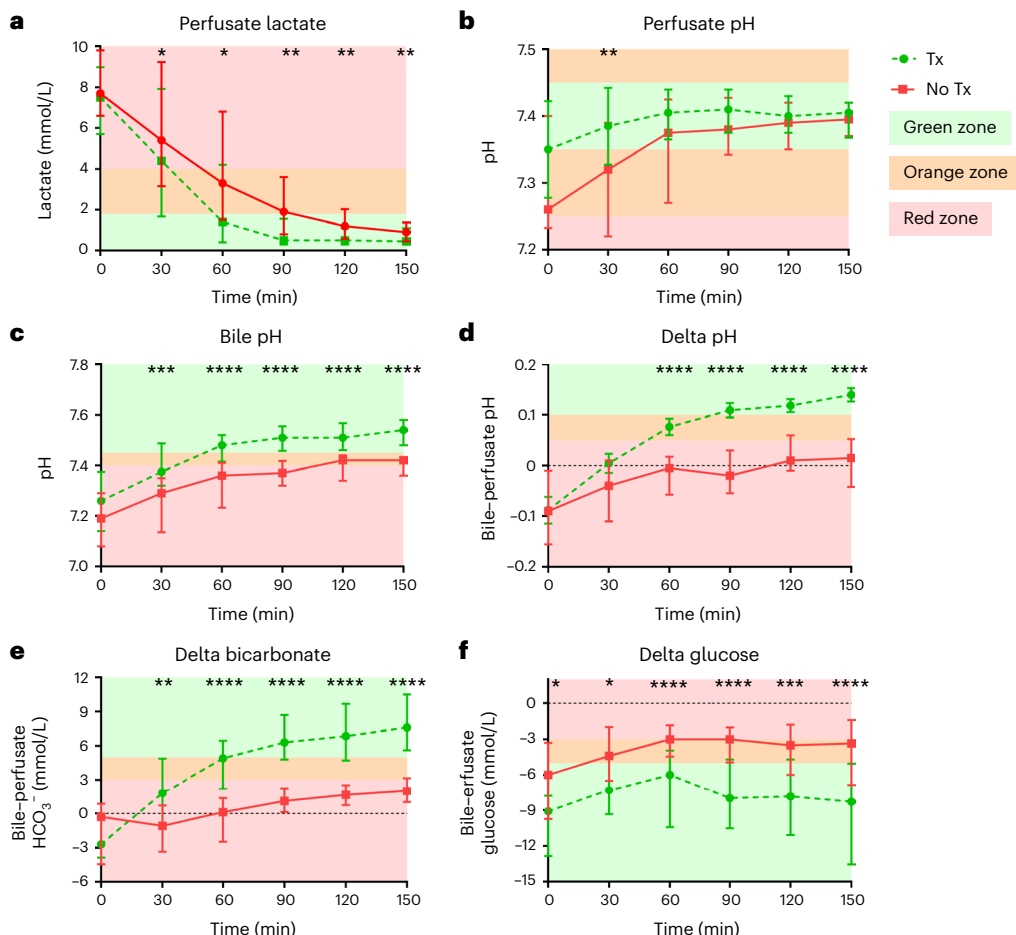


Fig. 5 | Viability criteria during normothermic machine perfusion. a,b, Biochemical parameters of the perfusate for hepatocellular function: perfusate lactate (a) and perfusate pH (b). c–f, Biochemical parameters of the bile for biliary function: bile pH (c), the delta of the perfusate pH and the bile pH (d), the delta between the perfusate bicarbonate and the bile bicarbonate (e), and the delta between the perfusate glucose and the bile glucose (f). The background color zones match the criteria indicated in Table 3. Data shown as median with 95% confidence intervals. Interestingly, nearly all livers met the hepatocellular function criteria. All organs that were declined for transplantation after DHOPE–COR–NMP did not meet the cholangiocellular viability criteria, while all hepatocellular criteria were met. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.00001$. Tx, transplanted livers; no Tx, not transplanted livers. Adapted with permission from ref. 26, Elsevier.

in the protocol. As nearly 100% of the livers met the hepatocellular viability criteria, one can assume that the utilization rate would be substantially higher if only high-risk donation after brain death livers (and not DCD livers) were used. So far, the livers subjected to the DHOPE–COR–NMP protocol have come mainly from older DCD donors, which are at an increased risk of ischemic biliary injury. However, recently reported outcomes from >100 procedures with DCD livers are excellent⁵⁰.

A distinction is made between the viability of the hepatocytes and the viability of the cholangiocytes. Hepatocyte function is assessed by the ability to clear lactate, maintain a stable pH and sufficient bile production^{14,15}. Several studies report the use of bile pH, bicarbonate and glucose as markers of bile duct viability^{16,31}. There are several additional markers that, together with the absolute pH value, provide a more complete indication of cholangiocyte function. These include the difference (‘deltas’) between bile and perfusate pH, bicarbonate and glucose. Values representing the biliary function of the transplanted and nontransplanted livers are shown in Fig. 5c–f. The development of the viability criteria is an ongoing process and refining of these criteria can only be performed by obtaining more experience with the procedure. Having performed over 200 DHOPE–COR–NMP procedures in Groningen and Rotterdam, we can conclude that it is safe to transplant livers that meet the criteria with this protocol. Our current utilization rate of livers after reconditioning and assessment during DHOPE–COR–NMP is ~70%. However, it could be that the criteria are still too strict and more livers could be transplanted. More research and collaboration are required to explore the boundaries.

All donor livers exposed to this protocol were initially declined based on donor characteristics or long functional warm ischemia time. However, the transplant outcomes of livers that met the viability criteria are comparable to benchmark studies, with an incidence of nonanastomotic biliary strictures of only 3% and 1 year graft survival of 94%^{26,51}. Therefore, our DHOPE–COR–NMP protocol will provide an important tool to reduce the high discard rate of ECD donor livers. We can safely increase the number of liver transplants with good transplant outcomes.

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O.B.v.L., V.A.L., V.E.d.M. and R.J.P. designed the protocol, wrote and critically revised the manuscript. B.L., A.M.T., S.B.B. and M.W.N. designed the protocol and critically revised the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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